ONLINE RETAIL

Sales Performance, Customer Behaviour & Data Quality Analysis



Project objective

Turn the Online Retail transaction dataset into a clear business view of sales performance, customer purchasing patterns, and data-quality exceptions using SQL and Power BI.

Prepared as a portfolio analysis using the SQL findings and project details supplied during the analysis. Dashboard pages in this report are static visual representations of those findings, not captures of a live Power BI file.

1. Executive Summary

This project analyses an online retail transaction dataset containing 541,909 raw rows. The analysis focuses on positive sales transactions, sales trends over time, geographic contribution, product descriptions, customer invoice frequency, and transaction-quality exceptions.

Metric	Result	Definition / note
Raw dataset rows	541,909	Original row count
Positive sales line value	10,666,684.54	Quantity > 0 and unit price > 0
Positive-sales invoices	19,960	Distinct invoice numbers in the positive-sales baseline
Identified customers	4,338	Distinct non-missing customer IDs in the positive-sales baseline
Average invoice value	534.40	Positive sales line value divided by positive-sales invoices
Positive sales lines	530,100	As reported in the supplied SQL results

The strongest headline is the high concentration of sales in the United Kingdom, which accounts for 84.61% of positive sales value. November 2011 is the highest month in the supplied monthly results. Customer invoice-frequency results show 2,010 identified customers with three or more invoices.

The analysis also identifies substantial missing customer identifiers and other transaction exceptions. These should be monitored and classified rather than silently removed, because different exception types can represent cancellations, returns, adjustments, or incomplete operational records.

2. Business Problem & Objective

Business problem. Transaction data can be difficult to use for decisions when sales performance, customer behavior, and data-quality issues are spread across raw rows and require separate queries. Without consistent definitions, users may mix positive sales with cancellations, returns, zero-price records, or incomplete customer details.

Objective. Establish a consistent positive-sales baseline, identify key sales and customer patterns, profile transaction exceptions, and present the findings in an accessible dashboard that supports investigation and decision-making.

Key questions

• How much positive sales value was recorded, and how many invoices and identified customers contributed?

• How did sales vary by month, country, weekday, and hour?

• Which product descriptions generated the highest sales value?

• How frequently did identified customers place invoices?

• What data-quality exceptions require monitoring or investigation?

3. Data & Method

The project uses the Online Retail transaction data loaded into PostgreSQL and analyzed through SQL. The cleaned analytical view used in Power BI is named **online_retail_clean**. The workflow includes loading the source data, profiling data quality, applying consistent transaction filters, answering business questions with SQL, and presenting the results in Power BI.

Technology	Role in the project
PostgreSQL / pgAdmin	Data storage, SQL analysis, aggregations, validation and business-question queries
Power BI	KPI cards, time trends, country and weekday comparisons, product/customer views, and exception monitoring
Report / presentation	Communicate findings, caveats, and recommended operational actions

Positive-sales definition. Unless otherwise stated, positive sales value is calculated from rows where Quantity > 0 and UnitPrice > 0. This keeps positive sales separate from negative quantity and zero-price records. Customer metrics use non-missing CustomerID values.

Interpretation boundary. Sales value is not profit. No cost, margin, inventory, or marketing-spend data has been supplied, so this report does not make profitability or return-on-investment claims.

4. Sales Performance

Monthly trend



Monthly positive sales value was highest in November 2011 at 1,509,496.33. September and October also show strong values. December 2011 records 638,792.68, lower than November; the available results do not establish whether this is due to seasonality, a partial reporting period, or a business change. Confirm the completeness of December 2011 before drawing a conclusion.

Geographic contribution

Country	Positive sales value	Share of total
United Kingdom	9,025,222.08	84.61%
Netherlands	285,446.34	2.68%
EIRE	283,453.96	2.66%
Germany	228,867.14	2.15%
France	209,715.11	1.97%

The UK is the dominant market in the supplied results. This concentration may create geographic dependency, but further analysis is required before recommending expansion or market investment. The next step would be to compare customer counts, average invoice value, repeat purchase, shipping costs, and margin by country.

Weekday and hourly patterns

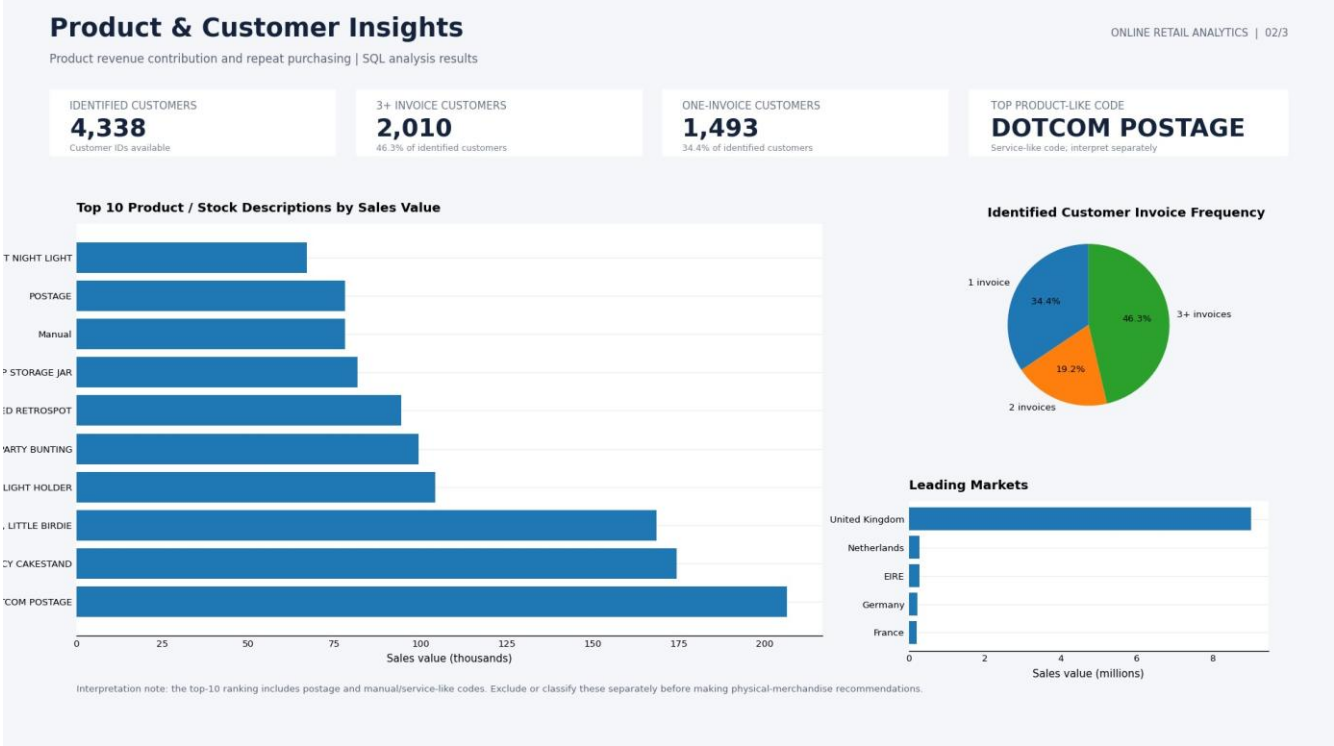
Weekday	Positive sales value
Thursday	2,203,161.24

Weekday	Positive sales value
Tuesday	2,178,632.61
Wednesday	1,851,147.81
Friday	1,840,340.23
Monday	1,779,575.04

Sunday	813,827.61

Thursday has the highest weekday sales value in the supplied results. Saturday was not shown in the output. The top sales hours were 10:00 (1,446,742.70) and 12:00 (1,444,245.88). These patterns can inform staffing or operational scheduling hypotheses but should be checked against transaction volumes and order-processing capacity.

5. Product & Customer Insights



Top product descriptions by sales value

Rank	Description	Sales value	Share
1	DOTCOM POSTAGE	206,348.77	1.93%
2	REGENCY CAKESTAND 3 TIER	174,484.74	1.64%
3	PAPER CRAFT, LITTLE BIRDIE	168,449.60	1.58%
4	WHITE HANGING HEART T-LIGHT HOLDER	104,340.29	0.98%
5	PARTY BUNTING	99,504.33	0.93%
6	JUMBO BAG RED RETROSPOT	94,340.65	0.88%
7	MEDIUM CERAMIC TOP STORAGE JAR	81,790.92	0.77%
8	Manual	78,110.27	0.73%
9	POSTAGE	78,101.88	0.73%
10	RABBIT NIGHT LIGHT	66,964.99	0.63%

The top-10 ranking includes postage and manual/service-like codes. These entries should be classified separately from physical merchandise before using the ranking to make product assortment or merchandising decisions.

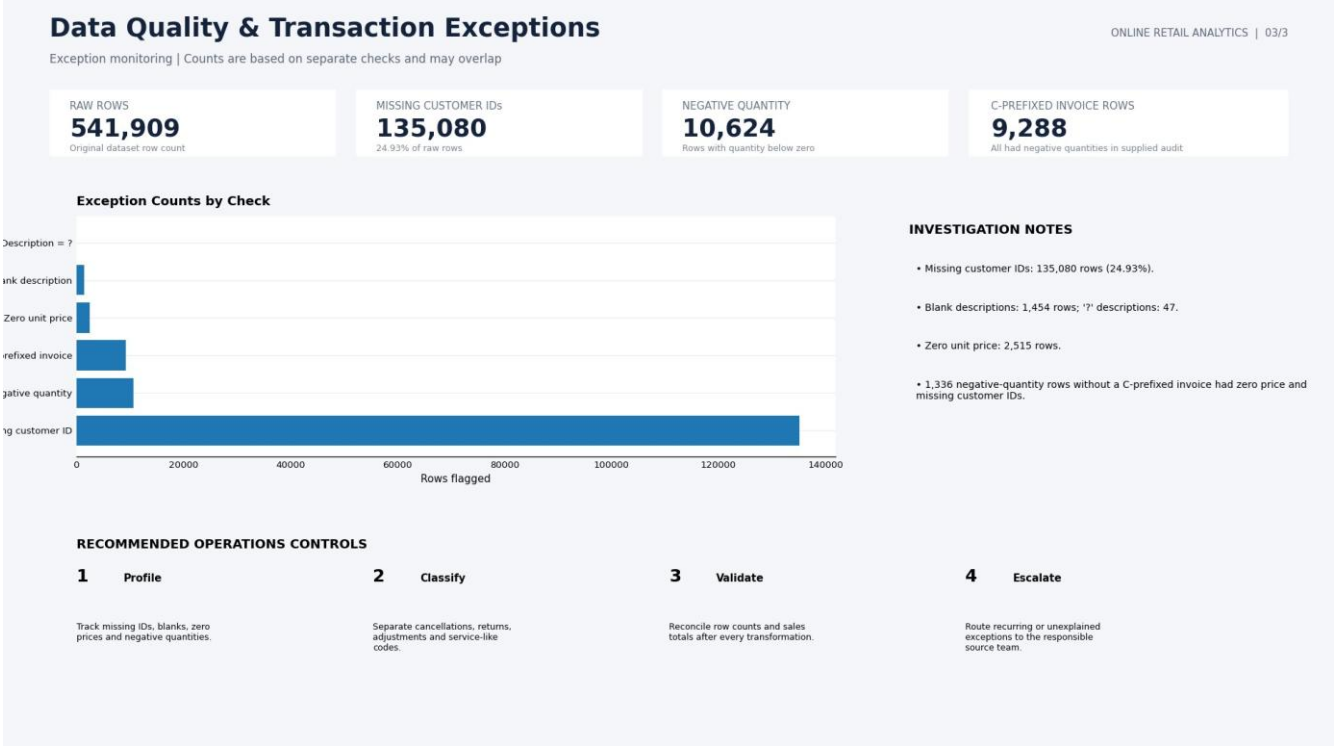
Customer invoice frequency

Invoice frequency	Identified customers	Share of 4,338
1 invoice	1,493	34.4%

2 invoices	835	19.2%
3+ invoices	2,010	46.3%

Nearly half of identified customers fall into the three-or-more-invoice group. This is a useful repeat-purchase signal, but it is not a formal retention or cohort analysis. Customers without an ID are excluded, and the analysis does not show time between purchases or customer lifetime value.

6. Data Quality & Transaction Exceptions



Exception check	Rows	Interpretation
Missing CustomerID	135,080	24.93% of all raw rows
Negative quantity	10,624	Negative-quantity transaction rows
C-prefixed invoice rows	9,288	All were negative-quantity rows in the supplied audit
Zero unit price	2,515	Requires classification and review
Blank description	1,454	Product-description completeness issue
Description = '?'	47	Uninformative description values
Negative quantity without C prefix	1,336	All had zero price, missing CustomerID and UK country in the supplied audit

The exception categories overlap and must not be summed to estimate unique affected rows. For example, a row can simultaneously have a missing customer ID and a negative quantity. The 1,336 negative-quantity rows without a C-prefixed invoice appear to include operational adjustment or damaged-goods descriptions; this is a hypothesis for investigation, not a confirmed business classification.

Recommended controls

- **Profile:** Track missing customer IDs, blank descriptions, zero prices, and negative quantities as recurring quality indicators.
- **Classify:** Separate cancellations, returns, operational adjustments, damaged goods, postage, and manual/service codes.
- **Validate:** Reconcile raw row counts and positive-sales totals after each transformation or refresh.

• **Investigation:** Review exception records with source teams before changing or excluding them.

• **Document:** Maintain definitions for positive sales, cancellations, customer counts, and product classifications.

7. Business Recommendations & Action Plan

Priority	Recommended action	Expected opera
1 — High	Create a recurring data-quality review for missing CustomerID, negative quantity, zero price and blank descriptions.	Improves visibili
2 — High	Standardize the positive-sales definition and keep cancellation/adjustment records separate in reporting.	Reduces incons
3 — High	Classify postage, manual, and service-like stock descriptions separately from merchandise.	Makes product
4 — Medium	Investigate UK sales concentration and compare markets using customer and order metrics.	Supports evidenc
5 — Medium	Validate the December 2011 reporting period and review monthly trend anomalies.	Prevents mislead
6 — Medium	Extend the model with cost, margin, inventory, and shipping data when available.	Enables profitabi

These are recommendations based on the patterns and limitations identified in the supplied results. The dataset alone does not prove the cause of missing IDs, negative quantities, or monthly changes.

8. Dashboard Guide

The accompanying dashboard previews are organized into three pages:

• **Sales Overview:** positive sales value, invoice count, identified customers, average invoice value, monthly trend, top countries, and weekday comparison.

• **Product & Customer Insights:** top product descriptions by sales value, customer invoice-frequency segments, and leading markets.

• **Data Quality & Exceptions:** raw row count, missing customer IDs, negative quantities, cancellation-prefixed rows, zero prices, blank descriptions, and recommended controls.

The dashboard images are static, report-ready visualizations generated from the SQL results supplied for this project. They are not screenshots exported from the user's interactive Power BI Desktop file. The live Power BI report should be refreshed and reconciled against the same definitions before publication.

9. Limitations

• No cost, gross margin, inventory, shipping cost, or marketing data was provided; therefore profitability and campaign effectiveness cannot be assessed.

• Customer analysis excludes rows with missing CustomerID and may not represent all purchasers.

• Product rankings include postage and manual/service-like descriptions; a separate merchandise classification is needed.

• Exception categories overlap and are not unique-row counts.

• The completeness of December 2011 is not confirmed.

- The supplied SQL outputs are the basis of the numerical findings; values should be revalidated against the final database refresh before external use.

10. Conclusion

The analysis provides a consistent view of positive sales performance, customer invoice frequency, geographic concentration, and transaction-quality exceptions. The most actionable findings are the UK concentration, the high November 2011 sales value, repeat invoice activity among identified customers, and the volume of missing customer identifiers. A practical next step is to operationalize the exception checks, standardize transaction classifications, and extend the dataset with cost and inventory fields to support deeper commercial decisions.